Brazilian researchers identify new species of microorganism in active volcano in Antarctica

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New species of archaeon was found in a fumarole, with temperatures close to 100°C, yet surrounded by ice and snow

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By: Luana Mendes*

Art by: Livia Bortoletto**

To reach this discovery, the researchers collected soil samples on Deception Island, in the South Pole – Photo: Courtesy of Amanda Bendia

A group of researchers from the Oceanographic Institute (IO) at USP discovered a new species of archaeon, a type of unicellular microorganism without a cell nucleus (prokaryote), in an active volcano in Antarctica. The genetic material of the microbe from the *Pyrodictiaceae* family was found in a sample collected from a fumarole – an opening in the ground through which hot gases of volcanic origin are released – on Deception Island, Antarctica. Using genetic analysis tools, the team reconstructed the archaeon's genome in order to identify characteristics related to its survival in such an extreme environment.

Amanda Bendia, professor at IO, works in microbial ecology and evolution in extreme marine environments, with a focus on the deep ocean and Antarctica. In 2014, the researcher took part in a scientific expedition to Antarctica under the Brazilian Antarctic Program aboard the polar vessel *Almirante Maximiano*. At the time, sediment samples from the fumarole were collected on Deception Island and sent for metagenomic sequencing, the analysis of genetic material from a complex sample without the need for prior cultivation, at the Brazilian laboratory.

At the time, the faculty member was a doctoral student at IO, supervised by professor Vivian Pellizari, a Brazilian pioneer in studies of microorganisms that live under extreme conditions. Years later, the sequenced genetic material was analyzed again, identifying a new genus and species of archaeon from the *Pyrodictiaceae* family. The new species was named *Pyroantarcticum pellizari*, in honor of microbiologist Vivian Pellizari.

Amanda Bendia - Photo: Lattes Curriculum

Ana Carolina Butarelli, doctoral student in microbiology at the Institute of Biomedical Sciences (ICB) at USP and researcher at IO's Microbial Ecology Laboratory (Lecom), and Francielli Vilela Peres, postdoctoral researcher in Biological Oceanography, also participated in the study.

The discovery was made possible through the MAG assembly technique – acronym for metagenome-assembled genome – in which a genome is reconstructed from sequencing data derived from environmental samples. Through various techniques involving computational systems, it is possible to obtain the genomes of organisms that cannot be cultivated in the laboratory, as is the case with some hyperthermophiles (which survive at temperatures above 60°C).

“Each organism present in the sample has a genome, and we often have millions of microorganisms in the material. So imagine having to segment and sequence the DNA to reconstruct the genome of these beings”, stated Ana Carolina Butarelli on the scale of the work.

Expedition carried out in 2014 aboard the polar vessel *Almirante Maximiano* took Brazilian researchers to Antarctica – Photo: Courtesy of the researchers

The discovery

The Archaea domain groups unicellular prokaryotic microorganisms, without a cell nucleus, morphologically similar to bacteria, but genetically and biochemically as distinct from them as they are from eukaryotes—which are organisms with nuclei in their cells, such as plants, fungi, algae and animals. The full formalization of the three-domain system, Bacteria, Archaea and Eukarya, was consolidated in the 1990s. In light of this, discoveries of archaeal species are recent, leading to major changes in science in a short period of time. “We are constantly discovering something new about archaea. *Pyrodictiaceae*, for example, was discovered about ten years ago”, said Ana Carolina Butarelli.

Antarctica currently has four active volcanoes, three on the continent and one on Deception Island. The continental volcanoes reach a maximum temperature of 65°C, which does not select for hyperthermophilic archaea; on the island, however, fumarole temperatures exceed 100°C. Previously, another group of hyperthermophilic archaea had been found by foreign researchers in fumaroles in Antarctica; however, *Pyrodictium*, a genus of the *Pyrodictiaceae* family, was located especially in hydrothermal vents of the deep ocean.

Scientific research in Antarctica is essential for understanding the global climate, serving as a natural laboratory for studies on climate change, biodiversity and astronomy – Photo: Amanda Bendia/Brazilian Academy of Sciences

Amanda Bendia explains that these hydrothermal vents reach temperatures above 400°C, in addition to providing chemical elements essential for sustaining the life of various microorganisms. These conditions contrast with the temperature of the surrounding water, which is around 4°C, typical of the deep sea. In this sense, the aforementioned factors associated with the atmospheric pressure of the environment indicate the survival of the species in extreme locations, but also suggests the hypothesis of the presence of mechanisms that enable transport through contrasting temperatures in the deep ocean or in Antarctica.

The scientists initially believed that the microorganism belonged to the same genus as the known group, because it inhabits deep-sea hydrothermal vents. However, the new archaeon lives in a surface fumarole, exposing a clear contrast with the temperature of the polar environment and also inhabiting an atmospheric pressure different from that of the other archaeal genus.

The new species belongs to one of the most primitive groups of living beings on the planet, known for living in extreme environments for billions of years – Photo: Courtesy of Amanda Bendia

Genomics and adaptations

To clearly identify the differences between these microorganisms, in order to classify a new genus and species, it is necessary to work with a protocol that addresses phylogeny, molecular adaptations, comparative genomics and the functions performed by these beings. Since it is not possible to cultivate them in the laboratory, due to the structural difficulty of simulating the organisms' living conditions, it is essential that the extracted genome be of the highest possible quality, with less than 10% contamination in the samples.

When working with genes recovered from an environmental sample, existing similarities in their composition are analyzed in order to indicate the phylogenetic relationship between the groups. This analysis can also indicate the metabolic activity of these beings, making it possible to infer their behaviors.

“When we access the genome, we have access to a snapshot of the genetic material, but we do not know whether that organism is actually transcribing and translating that material to produce a protein. However, we can infer that it has this ability, since that gene is within its genome”, noted Ana Carolina Butarelli.

Ana Carolina Butarelli - Photo: Lattes Curriculum

In addition, the characteristics identified in this genetic material demonstrate the adaptations of the molecules to their natural habitat. Proteins play a crucial role in the cell, acting as functional and structural agents, and it is possible to detect a phylogenetic relationship based on protein composition. *Pyroantarcticum pellizari*, the species discovered in the study, for example, has reverse gyrase – a protein that protects DNA from denaturing, that is, from having its strands separated at high temperatures – a common characteristic in hyperthermophilic archaea.

The analysis of unique genes in the newly assembled genome revealed a set of functionally distinct elements, including pathways for sulfur and nitrogen cycling, together with structural characteristics such as cannulae and stress-resistance systems, pointing to a survival strategy adapted to transient energy availability, metal stress and interactions at the community level in sediments. In this context, this genome offers an important overview of the potential of microbial life to thrive in multiple extremes, a topic that is increasingly relevant to studies in astrobiology, microbial bioprospecting and the impacts of climate change on polar ecosystems.

“When dealing with an organism that is not widely studied, or in our case, a new genus and species, having the complete genome directly affects the amount of information available about that organism. Therefore, the 97% genome purity rate is an important path for disseminating the discovery worldwide, as well as contributing to scientific databases”, highlighted Francielli Peres.

Achieving a high genome purity rate was not an easy task for the researchers, who took about a year to recover the DNA from the collected sample. In addition to the logistical obstacles of research on Deception Island, Francielli Peres said that the small number of studies available makes it difficult to gather information on the microorganisms. Laboratory and computational analyses were also obstacles, as they required major University infrastructure as well as the scientists' expertise. “Although our work may seem very glamorous, cool and incredible, there is also the complex side of being a scientist. Studying an organism that no one knows is an enormous challenge”, emphasized Ana Carolina Butarelli.

Francielli Vilela Peres - Photo: Courtesy of Francielli Vilela Peres

The study was presented by Ana Carolina Butarelli at ISME19, the *International Symposium on Microbial Ecology*, held in 2024 in Cape Town, South Africa – Photo: Courtesy of Ana Carolina Butarelli

Pyroantarcticum pellizari, the species discovered by the team, was submitted for official registration with SeqCode – a system of rules and nomenclature for Archaea and Bacteria based on genetic information. The name was evaluated and officially recognized. The researchers intend, in the future, to return to Deception Island and collect samples again from the fumarole, with the goal of attempting to cultivate the species in the laboratory.

The article *Hot life in Antarctica: a novel metabolically versatile Pyrodictiaceae genus thriving at a volcanic–cryosphere–marine interface* was published in the journal *ISME Communications* and can be accessed [**here**](https://academic.oup.com/ismecommun/advance-article/doi/10.1093/ismeco/ycag080/8550909).

Further information: amandagb@usp.br, with Amanda Gonçalves Bendia; anacarolinabutarelli@usp.br, with Ana Carolina de Araújo Butarelli; and francielliperes@usp.br, with Francielli Vilela Peres.

**Intern under the guidance of Fabiana Mariz*

***Intern under the guidance of Simone Gomes*

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